

Cell Design Specification — Power-Tool Battery Cell (final design V8)

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Values are mechanically taken from the parameter set / simulation outputs and annotated line by line; missing items are written "Not provided".

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 teaching parameterization, PyBaMM)	`--base Chen2020`
Rated capacity (Ah)	3.0 (design rating) / 2.6683 simulation-verified at 1C	bridge/params_v8.json `Nominal cell capacity [A.h]` / cell/r4_v8_1c_dfn.json `capacity_ah`
Voltage window (V)	2.5 – 4.2	base parameter set (lower/upper voltage cut-off)
Cell dimensions (mm)	electrodes 65 × 1580 unrolled; stack thickness 0.147 (50 pos + 60 neg + 9 sep + 16 Al + 12 Cu μm); shell: Not provided (no shell parameter)	base parameter set + params_v8.json + cell/r4_v8_energy.json `thickness_m`
Electrolyte formulation	baseline EC/EMC + LiPF6-class defaults; conductivity override $\sigma = 1.4 \text{ S/m}$; $t_{+} = 0.5$ (advanced concentrated-electrolyte class, estimate — marked)	params_v8.json `Electrolyte conductivity [S.m-1]`, `Cation transference number`
Thermal management	Total heat transfer coefficient $h = 45 \text{ W/m}^2\text{K}$ (ducted forced-air pack cooling; design value)	params_v8.json

2. Electrode and separator

Layer	Thickness (μm)	Porosity	AMVF	Particle radius (μm)	Density (kg/m ³)	Current collector
Positive (NMC811)	50	0.40	0.58	1.2	3262	Al 16 μm (2700 kg/m ³)
Negative (graphite)	60	0.42	0.56	2.0	1657	Cu 12 μm (8960 kg/m ³)
Separator	9	0.55	—	—	397	—

N/P ratio (negative electrode capacity density × thickness ÷ positive electrode capacity density × thickness, capacity density = $c_{\text{max}} \times F/3600 \times \text{AMVF}$ on electrode-volume basis):

$\text{N/P} = (33133 \times 0.56 \times 60\text{e-}6) / (63104 \times 0.58 \times 50\text{e-}6) = \mathbf{0.61}$ (full-lithiation caliber; c_{max} from base parameter set, thicknesses/AMVF from params_v8.json).

Design note: the cell is deliberately **anode-capacity-limited** — the 1C capacity scales with the negative AMVF (V6 2.9907 Ah × 0.56/0.63 = 2.658 vs V8 measured 2.6683 Ah), and the charged-state anode stoichiometry (29866/33133 = 0.9014 from the base-set initial concentration) leaves the anode below saturation at the 4C charge end, which is what yields the plating margin (see §5). The positive electrode cycles only in its low-stoichiometry region, so its excess full-lithiation capacity is inactive.

3. Process design parameters

Parameter	Value	Formula / note
Areal density, positive (g/m ²)	97.9	thickness × (1 – porosity) × density = 50e-6 × 0.60 × 3262
Areal density, negative (g/m ²)	57.7	60e-6 × 0.58 × 1657
Compaction density, positive (g/cm ³)	1.96	3262 × 0.60 / 1000
Compaction density, negative (g/cm ³)	0.96	1657 × 0.58 / 1000
Electrolyte fill amount (g)	6.18	pore volume 5.15e-6 m ³ × 1.2 g/cm ³ (literature value, annotated) × fill factor 1.0

Parameter	Value	Formula / note
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; production-line value requires tuning

4. Mass breakdown (g/cell)

Component	Mass (g)	Formula / source
Positive active material (NMC811)	9.715	$50\text{e-}6 \times 0.1027 \times 0.58 \times 3262$
Positive binder (PVDF, 1.5 vol%)	0.137	$50\text{e-}6 \times 0.1027 \times 0.015 \times 1780$ (PVDF density = literature default, no parameter in set)
Positive conductive additive (carbon black, 0.5 vol%)	0.057	$50\text{e-}6 \times 0.1027 \times 0.005 \times 2200$ (literature default)
Negative active material (graphite)	5.717	$60\text{e-}6 \times 0.1027 \times 0.56 \times 1657$
Negative binder (SBR/CMC, 2 vol%)	0.160	$60\text{e-}6 \times 0.1027 \times 0.02 \times 1300$ (literature default)
Separator	0.165	$9\text{e-}6 \times 0.1027 \times (1 - 0.55) \times 397$
Positive current collector (Al)	4.437	$16\text{e-}6 \times 0.1027 \times 2700$
Negative current collector (Cu)	11.042	$12\text{e-}6 \times 0.1027 \times 8960$
Electrolyte	6.180	pore volume $\times 1.2 \text{ g/cm}^3$ (excluded from contract mass — parameter set lacks electrolyte density)
Enclosure, tabs	Not modeled	outside pure-simulation boundary
Total (contract caliber)	**31.616**	$\Sigma \text{ layer thickness} \times \text{area} \times (1 - \text{porosity}) \times \text{density}$, electrolyte excluded (cell/r4_v8_energy.json `mass_kg`)
Total incl. electrolyte	37.796	$31.616 + 6.180$

BOM component sum (excl. electrolyte) 31.430 g vs contract 31.616 g: 0.6 % difference, binder/additive-density caliber (noted; no number altered).

5. Performance verification (vs entry-0 criteria, mechanical log-evaluate)

Metric	Value	Threshold	Verdict	Source
Nominal capacity (Ah)	2.6683	≥ 2.0	PASS	cell/r4_v8_1c_dfn.json:capacity_ah
5C discharge retention	0.9842	≥ 0.95	PASS	cell/r4_v8_5c_dfn.json ÷ cell/r4_v8_1c_dfn.json (cell/r4_v8_derived.json)
Power density (W/kg)	73121.8	≥ 4000	PASS	cell/r4_v8_energy.json:power_density_w_kg
T_max 5C discharge (K)	308.45	≤ 333.15	PASS	cell/r4_v8_5c_dfn.json:T_max_K
T_max 4C charge @45 °C (K)	323.57	≤ 333.15	PASS	cell/r4_v8_4c45_dfn.json:T_max_K
Plating on 4C charge	false (anode potential min +0.0119 V)	false	PASS	cell/r4_v8_4c45_dfn.json:anode_potential_v
Rated energy (Wh)	9.833	—	info	cell/r4_v8_energy.json:energy_wh
Energy density	311.0 Wh/kg / 651.3 Wh/L	—	info	cell/r4_v8_energy.json

6. Design notes (what changed vs baseline Chen2020 and why)

Particle radii: positive 5.22 $\rightarrow$ 1.2 μm (solid-diffusion time $\tau_D = r^2/D$ drops 6812 $\rightarrow$ 360 s — the root cause of the 8.7 % baseline 5C retention, fixed in round 2 and retained); negative 5.86 $\rightarrow$ 2.0 μm (interface surface for the 4C charge uniformity).

Electrode thickness 75.6/85.2 $\rightarrow$ 50/60 μm ; porosity 0.335/0.25 $\rightarrow$ 0.40/0.42; AMVF 0.665/0.75 $\rightarrow$ 0.58/0.56; separator 12 $\rightarrow$ 9 μm with porosity 0.47 $\rightarrow$ 0.55 — the rate/transport package that carries both the 5C retention and the plating margin (round 4, see log evaluate entries).

Electrolyte σ 0.95 $\rightarrow$ 1.4 S/m, t_{ion} 0.2594 $\rightarrow$ 0.5 — transport-class overrides (estimates, marked in propose entries).

Cooling h 10 $\rightarrow$ 45 W/m²K — thermal-management freedom used to hold $T_{\text{max}} \leq 333.15$ K.

Rated capacity 5.0 $\rightarrow$ 3.0 Ah: the reference set's 5.0 Ah belongs to the 75.6/85.2 μm electrode cell; the designed cell is rated at its own capacity so that 5C/4C protocols test the cell's own C-rates (probe-verified: cell/probe_nominal_1c_dfn.json). Rounds 1–3 had tested the thinned designs at effective 6.7–8.6C.

V8 vs alternatives: V7 (rated 3.6 Ah, 3.5321 Ah, retention 0.9775) is the higher-capacity alternative with a 10 mV plating margin; V8 was selected for the widest margins (retention 0.984, 4C thermal margin 9.6 K, plating margin 11.9 mV, 73.1 kW/kg).